

TABLE I
Free and open source software (FOSS)

Name	Architecture/OS	Description
FreePCB	Windows	A printed circuit board design program for Microsoft Windows. FreePCB allows up to 16 copper layers (metric and English units) and export of designs in Gerber format. Boards can be partially or fully auto-routed with the FreeRouting autorouter by using the FpcROUTE Specctra DSN file translator.
gEDA	FreeBSD, Mac, Linux	Suite of applications for schematic capture, PCB layout editing (with autorouting capability), Gerber viewing, analogue circuit simulation, Verilog design and GTK-based waveform viewing. Microsoft Windows-compatible binaries are no longer available.
KiCad	FreeBSD, Linux, Windows, Mac	Unlike other free software alternatives, KiCad provides the same interface for all design stages—schematic capture, PCB layout, Gerber generation/visualisation and library editing are all standard features. It also has a 3D view feature for PCBs. Migration tools (for transferring files from other EDA packages) are also provided.
Fritzing	Linux, Windows, Mac	Integrated tool for breadboard, schematic and PCB design. Targeted at non-engineers (designers, artists, researchers, hobbyists) and users of microcontroller platforms such as Arduino.
Verilator	Posix	Verilator is the fastest free Verilog HDL simulator. It compiles synthesisable Verilog into cycle-accurate C++ or SystemC code following two-state synthesis (zero-delay) semantics. It is reported that it is several times faster than commercial event-driven simulators such as ModelSim, NC-Verilog and VCS, while being not quite as fast as commercial cycle-accurate modeling tools such as Carbon ModelStudio and ARC VTOC.
Quite Universal Circuit Simulator	Linux, Solaris, Mac, NetBSD, FreeBSD, Windows	Schematic capture, Verilog, VHDL, simulation

zamiaCAD targets automation of active verification tasks and manual design, which increases verification engineer's productivity and register transfer level design.

Talking about hardware description language, the VHDL standard is completely supported on this software, but it may be extended by adding other frontends, e.g., Verilog.

Design entry support includes syntax highlighting, auto-completion and outlines. Under design navigation, instantiation graph on a fully elaborated model, open design unit through

instantiation graph, open design unit and open signal declaration via hierarchical paths can be done.

Simulation is performed with the help of a waveform viewer, source back-annotation of simulation values, waveform file import and a built-in simulator. The final design can be analysed with the help of reference and declaration search.

Some other free and open source software (FOSS)

The concept of open source and free sharing of technical information ex-

isted even before EDA tools came into the picture. The basic idea of open source software is that it should continuously evolve and develop itself from its initial release with the help of experiences and ideas coming from people using the software. With this kind of community cooperation, those experiences and ideas can be incorporated into future releases. There are a variety of other good open source EDA tools available. Compiled above is a list of them. ●

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EFY Group's New Initiative

EB TIMES

Launched on Sept 11, 2012



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